

Christopher R. Conner

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AREAS OF EXPERTISE

- Quantum technology researcher specializing in superconducting qubits and quantum acoustics
- Proficient at designing, simulating, fabricating, and measuring transmon qubit systems
- Author on more than 25 publications in peer-reviewed journals including 2 first-author papers
- Experienced science communicator with 5 conference presentations and countless informal presentations, as well as active collaboration with researchers from varied disciplines
- Mentorship of students in a variety of projects and settings

TECHNICAL SKILLS

- **Measurement:** Superconducting qubit and microwave resonator measurement via cryogenic low noise techniques utilizing custom DAC/ADC setups, QUICK boards, and vector network analyzers; also experienced with Signatone DC and microwave probe stations for device characterization
- **Fabrication:** Over 500 hours of experience in an ISO class 5 cleanroom, including both optical and ebeam lithography, as well as a variety of deposition and etching tools/techniques, Josephson junction deposition and characterization, air bridge fabrication, film characterization techniques such as X-ray diffraction, stylus profilometry, and atomic force microscopy, wirebonding, soldering, and machine shop mill and lathe usage
- **Cryogenics:** Extensive experience with Bluefors dilution refrigerator usage and maintenance, as well as HPD adiabatic demagnetization refrigerators
- **Software:** Sonnet, Ansys Q3D and HFSS, Comsol, QUCS, Python, QuTip, LabRAD, SolidWorks, AutoCAD, Git, LaTeX
- **Scientific Photography:** Macro-lense, microscope, and time-lapse photography; photo cover of September 25, 2023 issue of Applied Physics Letters and winner of 2022 Pritzker nanofabrication cleanroom image competition

EDUCATION

University of Chicago

2017 - Present

Pritzker School of Molecular Engineering, Chicago, IL

Ph.D. Candidate, Quantum Engineering, Expected Spring 2027

- PME Science Communications Fellow (2021-2022)
- IME GRA Fellowship (2017-2020)

University of California, Santa Barbara, Santa Barbara, CA

2013 - 2017

Bachelor of Science, Physics, June 2017

- Department of Physics Research Honors Award (2017)
- Department of Physics Academic Honors Award (2017)
- Letters and Science Honors Program (2013-2017)

RESEARCH EXPERIENCE

GRADUATE RESEARCH ASSISTANT,

2017 - Present

QUANTUM INFORMATION AND NANOMECHANICS LABORATORY

University of Chicago

Pritzker School of Molecular Engineering

Professor Andrew Cleland

Eckhart Research Center, Chicago, IL 60637

- Designed, simulated, fabricated, and measured a superconducting qubit experiment that culminated in randomized benchmarking of two-qubit entangling gates in a modular geometry; resulted in a first author publication in Applied Physics Letters and a presentation at APS March Meeting
- Collaborated with colleagues to perform measurements on a qubit-resonator system and develop analysis code used for Wigner tomography of exotic quantum states; presented the work at APS March Meeting

- Fabricated and measured high-Q acoustic microwave resonators with computation and sensing applications; currently writing a paper on these measurements and working towards quantum operation through integration with novel qubit designs; presented the work at APS March Meeting
- Coordinated planning and installation of multiple Bluefors dilution refrigerators and have an active role in their continued maintenance and operation
- Mentored multiple sets of undergraduate interns in various projects, with topics ranging from acoustic resonator design to microwave measurement techniques to software development for experiment control; also mentored graduate students in cleanroom fabrication techniques

UNDERGRADUATE RESEARCH ASSISTANT,
MICROFLUIDICS LABORATORY

2016 - 2017

University of California, Santa Barbara
Professor Sumita Pennathur

Engineering Sciences Building, UCSB, Santa Barbara, CA 93117

- Fabricated and measured a microfluidic realization of a Wimshurst energy converter; work resulted in a first author publication in *Physical Review Applied* and a presentation at the Southern California Micro and Nanofluidics Symposium
- Developed and led hands-on engineering education outreach activities at a local elementary school; communicated the lesson design in a presentation at the Southern California Micro and Nanofluidics Symposium

UNDERGRADUATE RESEARCH ASSISTANT,
PARTICLE PHYSICS LABORATORY

2016

Professor David Stuart

Broida Hall, UCSB, Santa Barbara, CA 93117

- Contributed to the development and assembly of a muon detector utilizing a three-dimensional grid of scintillating fibers and silicon photomultipliers

SELECTED PUBLICATIONS

1. R. Linehan, T. Trickle, **C. R. Conner**, et al. "Listening for new physics with quantum acoustics." *Physical Review D* 112, 115005 (2025). DOI: 10.1103/63zj-d8z4
2. **C. R. Conner**, A. Bienfait, H.-S. Chang, et al. "Superconducting qubits in a flip-chip architecture." *Applied Physics Letters* 118, 232602 (2021). DOI: 10.1063/5.0050173
 - Editor's pick
3. K. J. Satzinger, **C. R. Conner**, A. Bienfait, et al. "Simple non-galvanic flip-chip integration method for hybrid quantum systems." *Applied Physics Letters* 114, 173501 (2019). DOI: 10.1063/1.5089888
 - Featured article
4. **C. Conner**, T. de Visser, J. Loessberg, et al. "Energy Harvesting with a Liquid-Metal Microfluidic Influence Machine." *Physical Review Applied* 9, 044008 (2018). DOI: 10.1103/PhysRevApplied.0.044008

SELECTED CONFERENCE PRESENTATIONS

1. **C. Conner**, G. Andersson, H. Qiao, et al. (2026) "Lithium niobate HBARs with $Q > 1$ million compatible with superconducting Qubits." in the *American Physical Society March Meeting*, S16.00011, March 15-20, Denver, Colorado
2. M.-H. Chou, **C. Conner**, et al. (2024) "Deterministic multi-phonon entanglement between two mechanical resonators on separate chips." in the *American Physical Society March Meeting*, K53.00011, March 4-8, Minneapolis, Minnesota
3. **C. Conner**, K. Satzinger, Y. Zhong, et al. (2019) "Simple and Re-useable Flip-Chip Method for Hybrid Quantum Systems." in the *American Physical Society March Meeting*, L11.00010, March 4-8, Boston, Massachusetts
4. **C. Conner**, T. de Visser, J. Loessberg, et al. (2017) "Energy Harvesting with a Liquid-Metal Microfluidic Influence Machine." in the *Southern California Micro and Nanofluidics Symposium*, Aug. 25-26, Santa Barbara, California